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If the particle is thrown with initial velocity 'u' downward which is in negative y axis, then velocity and position at of the particle any time t is given by

$$v = u + gt \qquad (2.11)$$

$$y = ut + \frac{1}{2}gt^2 \qquad (2.12)$$

The square of the speed of the particle when it is at a distance y from the hill-top, is

$$v^2 = u^2 + 2gy \qquad (2.13)$$

Suppose the particle starts from rest.

Then u = 0

Then the velocity v, the position of the particle and v^2 at any time t are given by (for a point y from the hill-top)

$$v = gt \qquad (2.14)$$

$$y = \frac{1}{2}gt^2 \qquad (2.15)$$

$$v^2 = 2gy \qquad (2.16)$$

The time (t = T) taken by the particle to reach the ground (for which y = h), is given by using equation (2.15),

$$h = \frac{1}{2}gT^2 \qquad (2.17)$$

$$T = \sqrt{\frac{2h}{g}} \qquad (2.18)$$

The equation (2.18) implies that greater the height(h), particle takes more time(T) to reach the ground. For lesser height(h), it takes lesser time to reach the ground.

The speed of the particle when it reaches the ground (y = h) can be found using equation (2.16), we get

$$v_{ground} = \sqrt{2gh} \qquad (2.19)$$

The above equation implies that the body falling from greater height(h) will have higher velocity when it reaches the ground.

The motion of a body falling towards the Earth from a small altitude ($h << R$), purely under the force of gravity is called free fall. (Here R is radius of the Earth)

EXAMPLE 2.34

An iron ball and a feather are both falling from a height of 10 m.

a) What are the time taken by the iron ball and feather to reach the ground?
b) What are the velocities of iron ball and feather when they reach the ground? (Ignore air resistance and take g = 10 m s^{-2})

Solution

Since kinematic equations are independent of mass of the object, according to equation (2.8) the time taken by both iron ball and feather to reach the ground are the same. This is given by

$$T = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2\times10}{10}} = \sqrt{2}\,s \approx 1.414\,s$$

Thus, both feather and iron ball reach ground at the same time.